

# Multiplexing

## Frequency Division Multiplexing (FDM)



MUX

FDM.

DEMUX

Steps to Transmit

- (i) Request
- (ii) Path Establishment
- (iii) Transmit
- (iv) Call Disconnect.

Bandwidth  
(Max Freq - min Freq)



The Bandwidth of Carrier (Multiplexed)

$$= \sum_{i=1}^n \text{Carrier Frequency} + \sum_{i=1}^{n-1} \text{Guard Band.}$$

10  
 15  
 + 20 +  
 12  
 18  
 ---  
 75  
 + Guard Band

In FDM,  
Hierarchy of Multiplexing: →  
 Several voice channels are

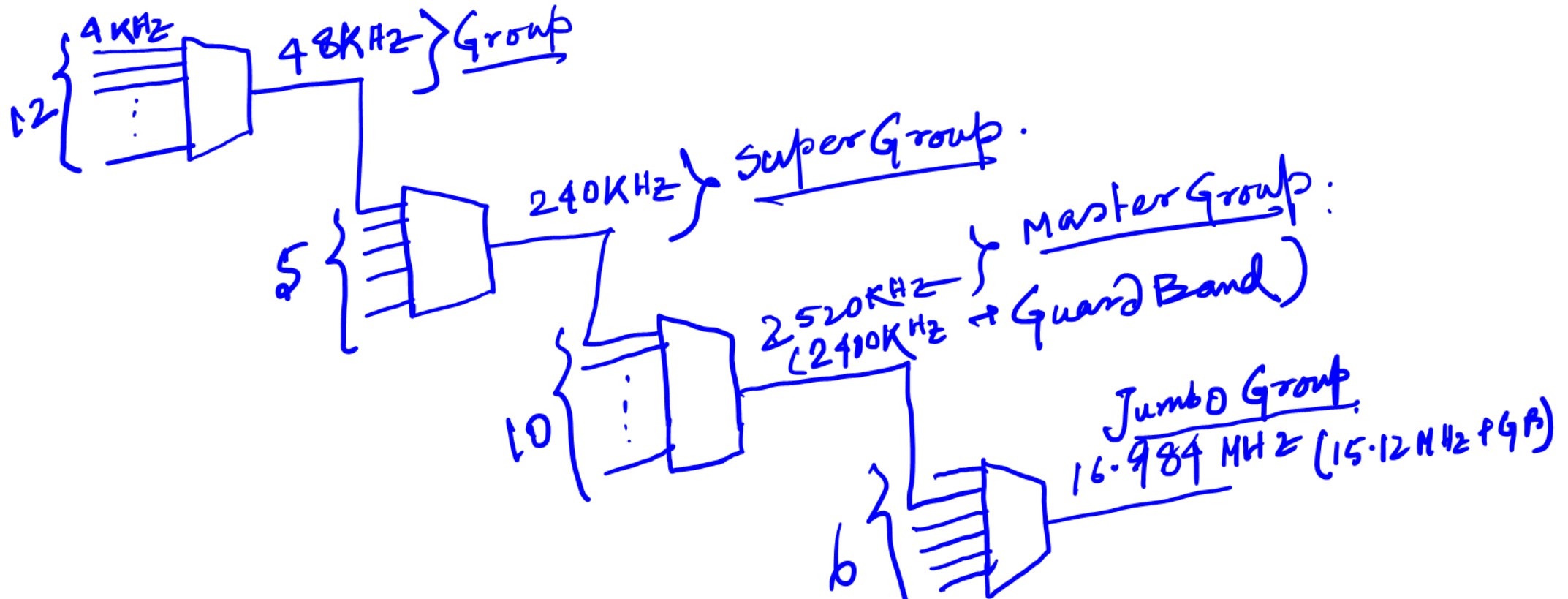
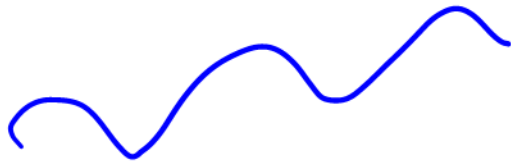


Fig. → Levels of Multiprogramming.

# Multiplexing

↓  
FDM  
Analog

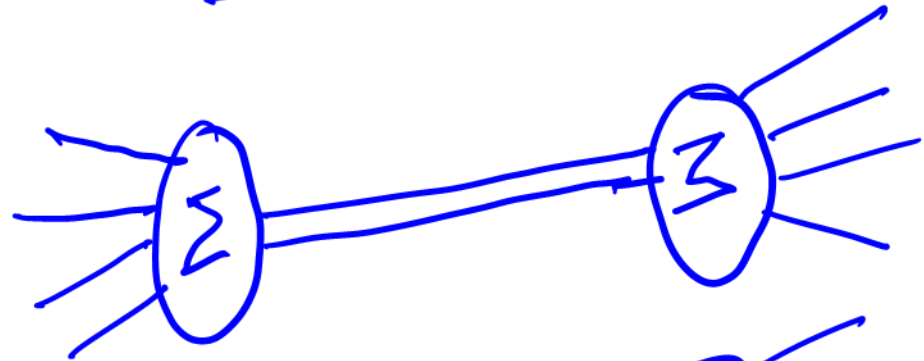
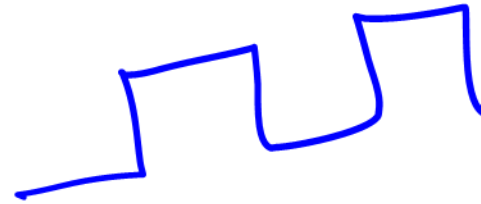


↓  
WDM  
Analog

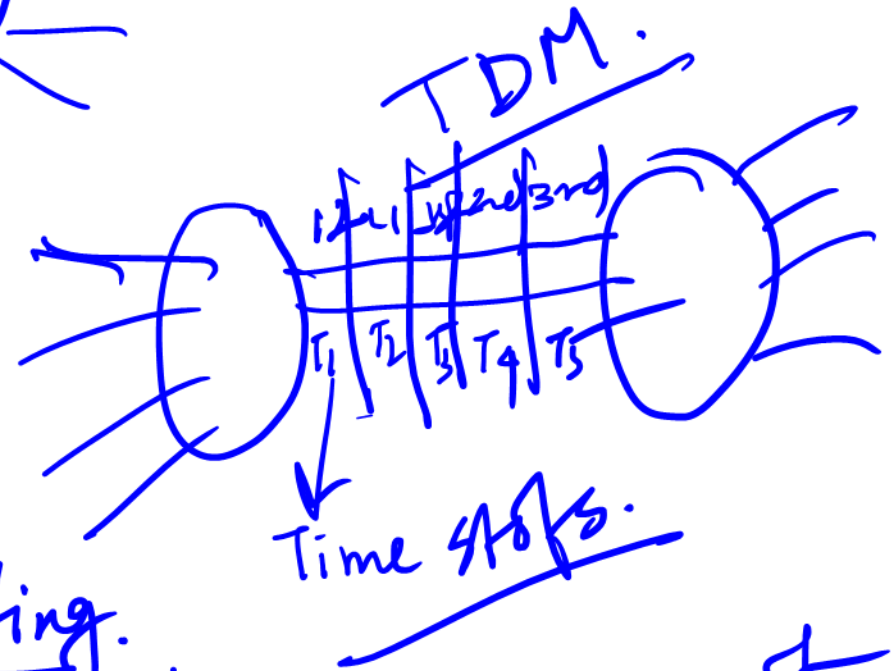
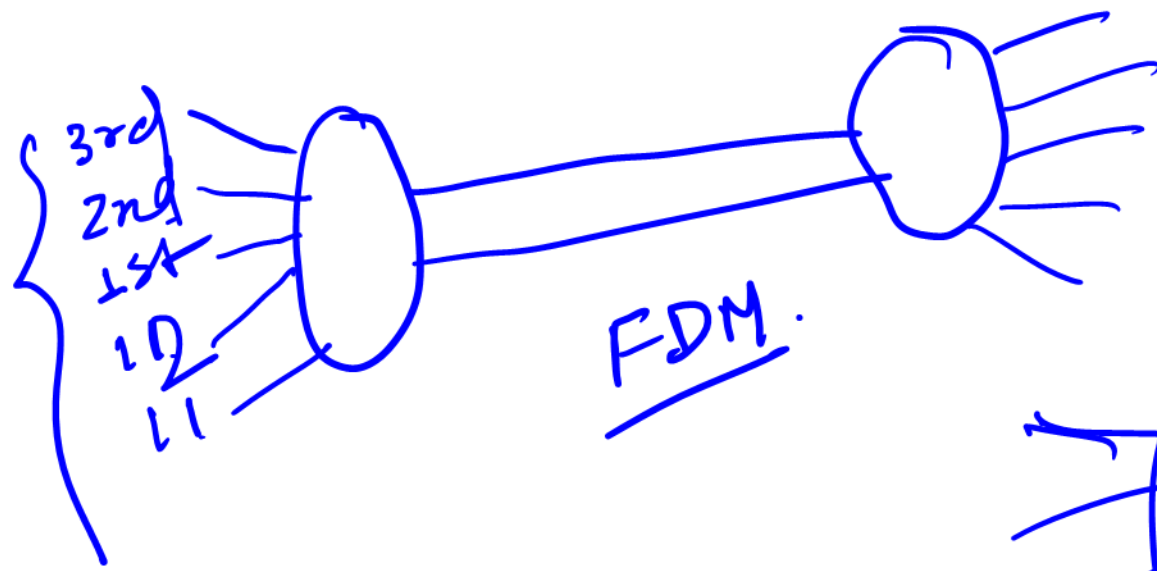


Time slot.

↓  
TDM.  
Digital.



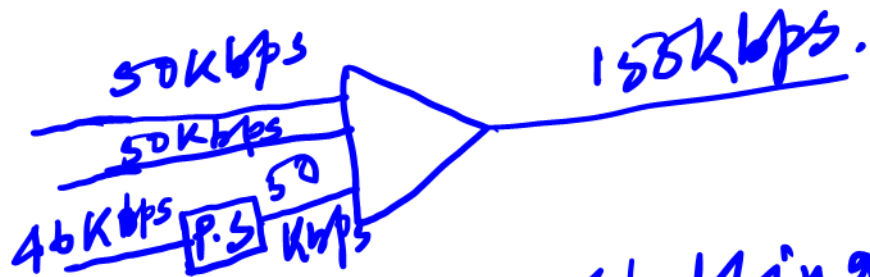
TDM.



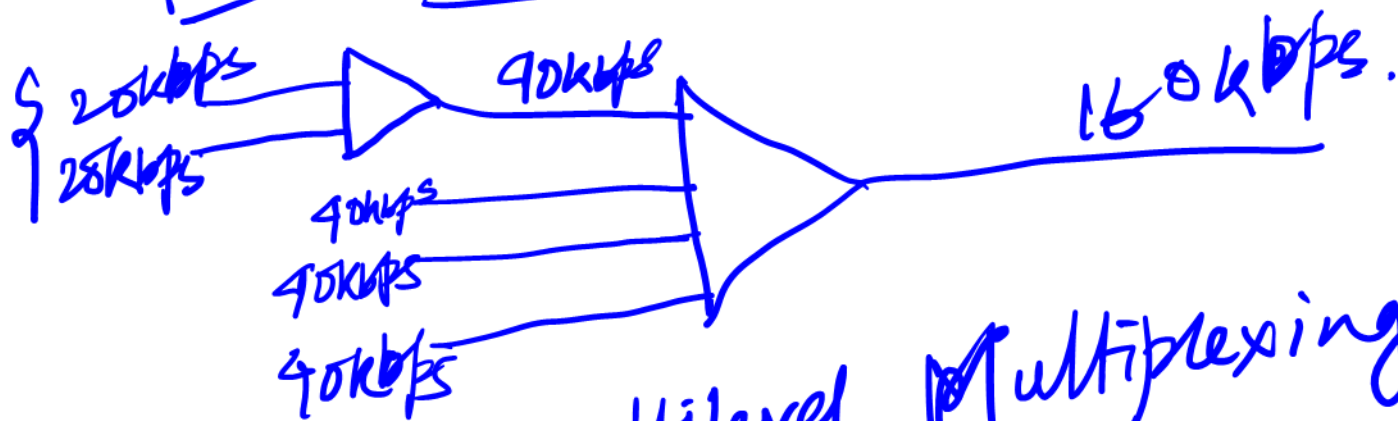
## Pulse stuffing / Bit Stuffing.

Sometimes the bit rate of source are not multiple integers of each other. Therefore, multilevel multiplexing and multiple slot allocation are not applied during transmission. To get rid of this problem, we make the highest input data rate (the dominant data rate) and

then add dummy bits to the input line with lower rate, this will increase the data rate. This technique is called pulse stuffing or bit stuffing.



Fig's Pulse stuffing.



Fig's Multilevel Multiplexing.

$$\begin{array}{r} 240 \\ - 038 \\ \hline \end{array}$$

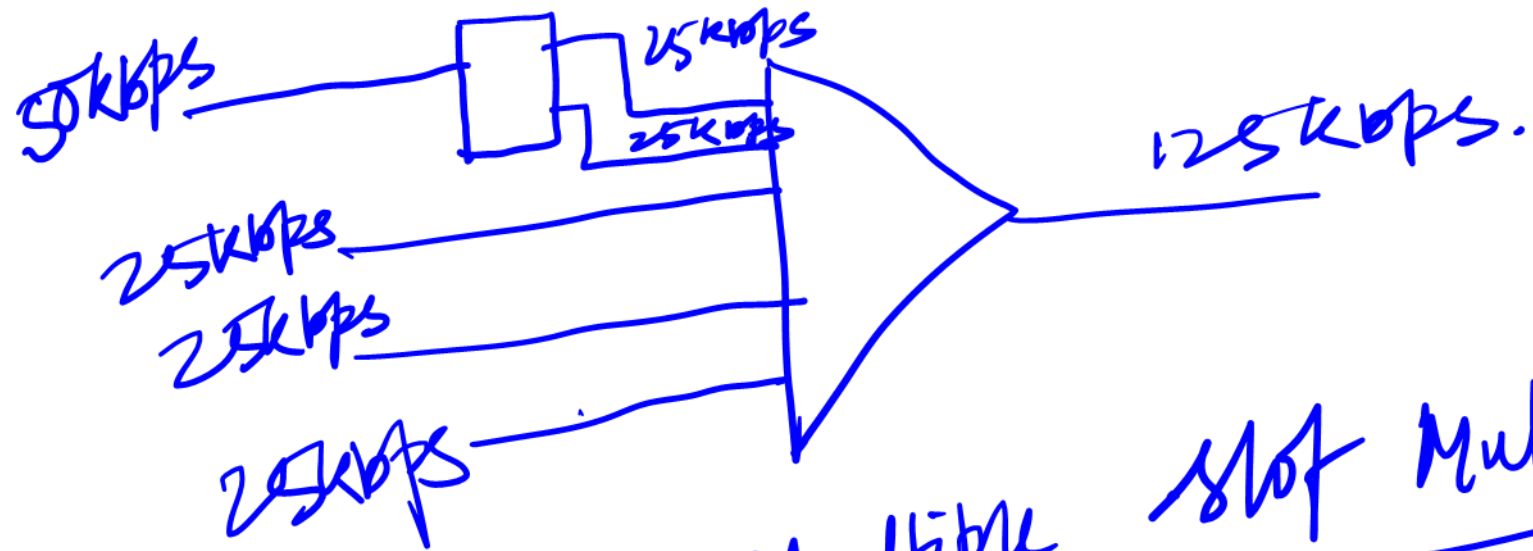


Fig:→ Multiple slot Multiplexing.



